

International Cybersecurity and Digital Forensics Academy (ICDFA)

SBT-DF203: Basic Networking Skills for Digital Forensics

LAB 7: DNS Introduction and Traffic Analysis

Three-Week Delivery Block 3/3 | 19-25 September 2026

Prepared by: Aminu Idris, AMCPN | Founder, ICDFA

Important Pre-Lab Instruction Read the matching lecture slide deck before executing the practical. Work only inside the ICDFA-approved lab or your own isolated virtual environment. Record every command, observation and screenshot as contemporaneous evidence.	
Course Code	SBT-DF203
Course Title	Basic Networking Skills for Digital Forensics
Lab Number	Lab 7
Lab Title	DNS Introduction and Traffic Analysis
Delivery Block	3/3 of 3
Scheduled Dates	19-25 September 2026
Estimated Duration	3-4 hours practical work plus report writing
Mode	Individual practical lab in an authorized virtual environment
Required Evidence	dig_dns.pcap or locally generated DNS captures; optional smtp.pcap correlation

1. Source Materials and Slide Alignment

- Domain Name System Forensics: domain-to-IP resolution, hierarchical DNS, resolver/TLD/authoritative roles, dig queries, capture with tshark, query/response fields and DNS within email traffic.
- The slide practical captures dig and browser DNS traffic, examines A records, local resolver configuration, TTL and authoritative information.
- The exercise builds the normal DNS baseline required before analyzing spoofing in Lab 8.

Reading Requirement Before Lab Execution

The lab deliberately follows the sequence, terminology and core exercises presented in the uploaded SBT-DF203 slides. Commands have been corrected where needed for Linux syntax and forensic repeatability.

2. Lab Scenario

You are documenting normal DNS behaviour for an incident-response baseline. You will identify the configured resolver, capture controlled dig queries, analyze transaction IDs, names, record types, response codes, answers and TTL values, and correlate DNS resolution with a subsequent web or email connection.

3. Learning Outcomes

- Explain recursive and iterative DNS resolution at a foundational level.
- Use dig to query A, AAAA, MX and NS records.
- Capture and analyze DNS queries and responses with tshark/Wireshark.
- Identify transaction ID, query name, record type, response code, answer and TTL.
- Correlate DNS responses with later IP connections.
- Recognize normal variations such as multiple answers, CNAMEs, cache responses and IPv6 queries.

4. Legal, Ethical and Safety Rules

- Use only systems, packet captures and networks for which you have explicit authorization.
- Keep the lab isolated from production, campus, office and public networks. Use NAT or host-only virtual networking as instructed.
- Do not collect, disclose or reuse real credentials, personal messages or confidential traffic.
- Preserve original capture files. Perform analysis on verified working copies and record SHA-256 hashes.
- Stop any traffic-generation or interception process immediately after the required evidence is captured.
- Restore firewall, ARP, forwarding and network settings before closing the lab.

5. Required Tools and Environment

Item	Recommended Tool	Purpose
Operating system	Kali Linux or Ubuntu VM	Generate and analyze DNS traffic.
DNS utility	dig from dnsutils	Perform controlled queries.
Capture tools	tshark and Wireshark	Capture DNS traffic.
Evidence	dig_dns.pcap or new PCAPNG	Analysis source.
Resolver config	/etc/resolv.conf, resolvectl	Identify configured DNS service.
Optional browser	Firefox/Chrome	Generate multiple DNS lookups.

6. Lab Folder Structure and Evidence Preparation

```
mkdir -p ~/SBT-DF203-Lab7/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab7
pwd
find . -maxdepth 1 -type d -print
```

Create the folder structure before downloading or generating evidence. Store original captures under evidence and analysis copies under working.

```
sudo apt update
sudo apt install -y dnsutils tshark wireshark
cat /etc/resolv.conf | tee reports/resolv_conf.txt
resolvectl status 2>/dev/null | tee reports/resolvectl_status.txt || true
wget -O evidence/dig_dns.pcap \
  'https://raw.githubusercontent.com/frankwxu/digital-forensics-
  lab/main/Networking_Forensics/lab_files/dns/dig_dns.pcap'
cp --preserve=timestamps evidence/dig_dns.pcap working/dig_dns_working.pcap
sha256sum evidence/dig_dns.pcap working/dig_dns_working.pcap | tee reports/dns_capture_hashes.txt
```

```

Session Actions Edit View Help
(kali@kali)-[~]
$ mkdir -p ~/SBT-DF203-Lab7/{evidence,working,exported,reports,screenshots,scripts}

(kali@kali)-[~]
$ cd ~/SBT-DF203-Lab7

(kali@kali)-[~/SBT-DF203-Lab7]
$ pwd
/home/kali/SBT-DF203-Lab7

(kali@kali)-[~/SBT-DF203-Lab7]
$ find . -maxdepth 1 -type d -print
.
./exported
./evidence
./working
./screenshots
./scripts
./reports

(kali@kali)-[~/SBT-DF203-Lab7]
$ sudo apt install -y dnsutils tshark wireshark
[sudo] password for kali:
Note, selecting 'bind9-dnsutils' instead of 'dnsutils'
tshark is already the newest version (4.6.6-1).
wireshark is already the newest version (4.6.6-1).
The following packages were automatically installed and are no longer required:
gdal-data libcrypt-dev libhdf4-0 libmbcrypto16 libproj25 libzxing3 python3.13-minimal
gdal-plugins libfreeexl1 libhdf5-310 libmjpegutils-2.1-0t64 libpython3.13-dev libzboard-common python3.13-venv
gir1.2-ayatanaappindicator3-0.1 libfyba0t64 libhdf5-hl-310 libmpeg2encpp-2.1-0t64 libraw23t64 proj-bin rpcsvc-proto
gir1.2-girepository-2.0 libgav1-1 libjs-jquery-ui libmpeg2-2.1-0t64 librttopo1 proj-data samba-ad-provision
libaec0 libgdal37 libjs-underscore libmuparser2v5 libspatialite8t64 python3-fs samba-dsdb-modules
libarmadillo14 libgeos-c1t64 libjxr-tools libnetcdf22 libwireshark18 python3-gpg tdb-tools
libarpack2t64 libgeos3.14.0 libjxr0t64 libodbc2 libwiretap15 python3-gpg unixodbc-common
libavc1394-0 libgeotiff5 libkmlbase1t64 libodbcinst2 libwmflite-0.2-7 python3-tdb python3.13
libblosc2-4 libgirepository-1.0-1 libkmlcore1t64 libwsutil16 python3.13
libcfitsio10t64 libpgpme45 libkmlengine1t64 libportmidi0 libxerces-c3.2t64 python3.13-dev

Use 'sudo apt autoremove' to remove them.

Upgrading:
bind9-dnsutils bind9-host bind9-lb

```

```

(kali@kali)-[~/SBT-DF203-Lab7]
$ cat /etc/resolv.conf | tee reports/resolv_conf.txt
# Generated by NetworkManager
search localdomain
nameserver 192.168.225.2
nameserver 192.168.88.1

(kali@kali)-[~/SBT-DF203-Lab7]
$ resolvectl status 2>/dev/null | tee reports/resolvectl_status.txt || true
Do you want to install it? (N/y)y
Reading package lists ...
Building dependency tree ...
Reading state information ...
The following packages were automatically installed and are no longer required:
gdal-data libjs-jquery-ui libwiretap15
gdal-plugins libjs-underscore libwmflite-0.2-7
gir1.2-ayatanaappindicator3-0.1 libjxr-tools libwsutil16
gir1.2-girepository-2.0 libjxr0t64 libxerces-c3.2t64
libaec0 libkmlbase1t64 libzboard-common
libarmadillo14 libkmlcore1t64 libzboard-common
libarpack2t64 libkmlengine1t64 proj-bin
libavc1394-0 libmbcrypto16 proj-data
libblosc2-4 libmjpegutils-2.1-0t64 python3-fs
libcfitsio10t64 libmpeg2encpp-2.1-0t64 python3-gpg
libcrypt-dev libmpeg2-2.1-0t64 python3-gpg
libfreeexl1 libmuparser2v5 python3-talloc
libfyba0t64 libnetcdf22 python3-tdb
libgav1-1 libodbc2 python3.13
libgdal37 libodbcinst2 python3.13-dev
libgeos-c1t64 libportmidi0 python3.13-minimal
libgeos3.14.0 libproj25 python3.13-venv
libgeotiff5 libpython3.13-dev rpcsvc-proto
libgirepository-1.0-1 libraw23t64 samba-ad-provision
libpgpme45 librttopo1 samba-dsdb-modules
libhdf4-0 libspatialite8t64 tdb-tools
libhdf5-310 libwireshark18 unixodbc-common
libhdf5-hl-310

Use 'sudo apt autoremove' to remove them.

```

```

(kali@kali)~[~/SBT-DF203-Lab7]
$ sudo cp /etc/resolv.conf /etc/resolv.conf.backup

(kali@kali)~[~/SBT-DF203-Lab7]
$ echo "nameserver 8.8.8.8" | sudo tee /etc/resolv.conf
nameserver 8.8.8.8

(kali@kali)~[~/SBT-DF203-Lab7]
$ cat /etc/resolv.conf
nameserver 8.8.8.8

(kali@kali)~[~/SBT-DF203-Lab7]
$ wget -O evidence/dig_dns.pcap \
'https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Networking_Forensics/lab_files/dns/dig_dns.pcap'
--2026-09-22 22:19:52-- https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Networking_Forensics/lab_files/dns/dig_dns.pcap
Resolving raw.githubusercontent.com (raw.githubusercontent.com) ... 185.199.111.133, 2606:50c0:8001::154
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.111.133|:443 ... connected.
HTTP request sent, awaiting response ... 200 OK
Length: 1804656 (1.7M) [application/octet-stream]
Saving to: 'evidence/dig_dns.pcap'

evidence/dig_dns.pcap      100%[=====>]  1.72M  203KB/s   in 9.3s

2026-09-22 22:20:02 (189 KB/s) - 'evidence/dig_dns.pcap' saved [1804656/1804656]

(kali@kali)~[~/SBT-DF203-Lab7]
$ cp --preserve=timestamps evidence/dig_dns.pcap working/dig_dns_working.pcap

(kali@kali)~[~/SBT-DF203-Lab7]
$ sha256sum evidence/dig_dns.pcap working/dig_dns_working.pcap | tee reports/dns_capture_hashes.txt
9a7c1b95aa6d60f8ac0ee9a79e06ae51145bb7768974526546316cd3dc91375d  evidence/dig_dns.pcap
9a7c1b95aa6d60f8ac0ee9a79e06ae51145bb7768974526546316cd3dc91375d  working/dig_dns_working.pcap

(kali@kali)~[~/SBT-DF203-Lab7]
$ sha256sum evidence/dig_dns.pcap working/dig_dns_working.pcap
9a7c1b95aa6d60f8ac0ee9a79e06ae51145bb7768974526546316cd3dc91375d  evidence/dig_dns.pcap
9a7c1b95aa6d60f8ac0ee9a79e06ae51145bb7768974526546316cd3dc91375d  working/dig_dns_working.pcap

(kali@kali)~[~/SBT-DF203-Lab7]

```

7. Mini Evidence and Chain-of-Custody Worksheet

Field	Student Entry
Case/lab identifier	SBT-DF203-Lab7-Samuel_Chinedu_Onu
Trainee name	Samuel Chinedu Onu
Date and time started	22 nd Sept 2026, 10am
Evidence file name(s)	dig_dns.pcap
Source or generation method	Downloaded from GitHub (frankwxu digital-forensics-lab repository)
Original SHA-256	9a7c1b95aa6d60f8ac0ee9a79e06ae51145bb7768974526546316cd3dc91375d
Working-copy SHA-256	9a7c1b95aa6d60f8ac0ee9a79e06ae51145bb7768974526546316cd3dc91375d
Analysis workstation/VM	Kali Linux VM
Notes on any changes	Temporary DNS change to 8.8.8.8 was made so the download could succeed. Original resolv.conf was backed up.

8. Part A - Query DNS Records with dig

```

dig example.com A | tee reports/dig_example_A.txt
dig example.com AAAA | tee reports/dig_example_AAAA.txt
dig example.com MX | tee reports/dig_example_MX.txt
dig example.com NS | tee reports/dig_example_NS.txt
dig +short example.com A | tee reports/dig_example_short.txt

```

- Record the resolver shown in the SERVER line.
- Record status/response code, flags, question count, answer count and query time.
- Explain IN, A, AAAA, MX and NS.
- Record at least one TTL and explain that it is a cache lifetime, not a creation date.

```
Session  Actions  Edit  View  Help
└─$ dig example.com A | tee reports/dig_example_A.txt

; <<>> DiG 9.20.27-2-Debian <<>> example.com A
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 21903
;; flags: qr rd ad; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 0
;; WARNING: recursion requested but not available

;; QUESTION SECTION:
;example.com.                IN      A

;; ANSWER SECTION:
example.com.                 9085    IN      A      172.66.147.243

;; Query time: 31 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Tue Sep 22 22:33:27 WAT 2026
;; MSG SIZE  rcvd: 45

(kali㉿kali)-[~/SBT-DF203-Lab7]
└─$ dig example.com AAAA | tee reports/dig_example_AAAA.txt

; <<>> DiG 9.20.27-2-Debian <<>> example.com AAAA
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 58926
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags;; udp: 512
;; QUESTION SECTION:
;example.com.                IN      AAAA

;; ANSWER SECTION:
example.com.                 300     IN      AAAA    2606:4700:10::6814:179a
example.com.                 300     IN      AAAA    2606:4700:10::ac42:93f3

;; Query time: 331 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Tue Sep 22 22:34:53 WAT 2026
;; MSG SIZE  rcvd: 96
```

```

Session Actions Edit View Help
(kali㉿kali)-[~/SBT-DF203-Lab7]
$ dig example.com MX | tee reports/dig_example_MX.txt

; <<>> DiG 9.20.27-2-Debian <<>> example.com MX
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 60873
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;example.com.                IN      MX

;; ANSWER SECTION:
example.com.                128     IN      MX      0 .

;; Query time: 213 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Tue Sep 22 22:36:37 WAT 2026
;; MSG SIZE rcvd: 55

(kali㉿kali)-[~/SBT-DF203-Lab7]
$ dig example.com NS | tee reports/dig_example_NS.txt

; <<>> DiG 9.20.27-2-Debian <<>> example.com NS
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 16563
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;example.com.                IN      NS

;; ANSWER SECTION:
example.com.                21600   IN      NS      elliottns.cloudflare.com.
example.com.                21600   IN      NS      herans.cloudflare.com.

;; Query time: 276 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Tue Sep 22 22:38:52 WAT 2026
;; MSG SIZE rcvd: 95

```

```

(kali㉿kali)-[~/SBT-DF203-Lab7]
$ dig +short example.com A | tee reports/dig_example_short.txt
104.20.23.154

(kali㉿kali)-[~/SBT-DF203-Lab7]
$

```

9. Part B - Capture a Fresh DNS Query

```

IFACE=eth0
sudo tshark -i "$IFACE" -f 'port 53' -a duration:25 -w evidence/fresh_dig_dns.pcapng &
sleep 3
dig +noedns example.com A >/dev/null
wait
sha256sum evidence/fresh_dig_dns.pcapng | tee reports/fresh_dns_sha256.txt

```

```
kali@kali: ~/SBT-DF203-Lab7
Session Actions Edit View Help

(kali@kali)~[~/SBT-DF203-Lab7]
$ IFACE=eth0

(kali@kali)~[~/SBT-DF203-Lab7]
$ sudo tshark -i "$IFACE" -f 'port 53' -a duration:25 -w evidence/fresh_dig_dns.pcapng & sleep 3

[4] 32069
[sudo] password for kali:
[4] + suspended (tty input) sudo tshark -i "$IFACE" -f 'port 53' -a duration:25 -w

(kali@kali)~[~/SBT-DF203-Lab7]
$ dig +noedns example.com A >/dev/null

(kali@kali)~[~/SBT-DF203-Lab7]
$ wait

(kali@kali)~[~/SBT-DF203-Lab7]
$ sha256sum evidence/fresh_dig_dns.pcapng | tee reports/fresh_dns_sha256.txt
sha256sum: evidence/fresh_dig_dns.pcapng: No such file or directory
```

Interface Note

On systems using a local stub resolver, packets may appear on lo and then on a physical interface. Use tshark -D to list interfaces and choose the one that shows the actual query.